Do Information Inputs Causally Affect LLM-Generated Investment Portfolios?

A Randomized Experiment on Information Conditioning and Allocation Behavior

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Abstract

Large language models (LLMs) are increasingly asked to produce structured financial recommendations, yet the degree to which their outputs are shaped by *which* information is placed in the prompt — as opposed to the model's own priors — remains unclear. I run a randomized experiment in which Anthropic's `claude-sonnet-4-6` is asked to allocate capital across a fixed universe of 20 U.S. large-cap stocks under four prompt conditions: (i) a *control* prompt with no financial data; (ii) a *fundamental* prompt containing market cap, P/E, revenue and earnings growth, and leverage; (iii) a *technical* prompt containing returns, volatility, beta, and moving-average signals; and (iv) a *combined* prompt containing both. Each condition is run ∼30 times at temperature 0.7, yielding 119 valid portfolios. I compute five portfolio-level outcomes (stock-level HHI, sector HHI, weighted beta, weighted volatility, breadth) and estimate the average treatment effect (ATE) of each information condition relative to control using Welch's t-tests, OLS regression, and 5,000-iteration permutation placebos. Effects are large and highly significant: the fundamental prompt increases stock-level concentration (HHI) by +0.043 (a 72 % relative increase, $p < 0.001$) and sector-level concentration by +0.157, whereas the technical prompt pushes the portfolio toward lower-beta, lower-volatility allocations ($\Delta\beta = -0.14$, $p < 0.001$). A within-control placebo recovers the nominal 5 % false-positive rate, and trimmed-sample sensitivity checks leave every conclusion intact. The experiment demonstrates that information framing is a causal driver of LLM portfolio behavior that is practically detectable with modest sample sizes.

1 Introduction

LLMs are being embedded into investment research and client-facing advisory workflows at a pace that has outrun the scholarly literature on their behavior under different information regimes. A vendor building an "AI analyst" makes an implicit causal claim every time it argues that its product produces better recommendations because it is given better information. Testing that claim requires treating the information in the prompt as a *manipulable treatment* and the LLM's output as a *response variable* — which is a natural setting for a randomized experiment.

This project asks a simple causal question:

> *Holding the model, temperature, universe, and instruction template fixed, does the type of financial information provided in the prompt causally change the portfolio an LLM constructs?*

I answer it with a fully randomized 4-arm experiment. The four arms differ only in the data block included in the prompt: nothing, fundamentals, technicals, or both. The outcome is a portfolio of weights across 20 stocks, summarized by five well-known portfolio-level metrics. Because assignment to condition is randomized at the trial level and everything else is held fixed, any systematic differences in the mean outcomes are attributable to the information injected into the prompt.

The contribution is threefold. First, I design a clean causal experiment that cleanly isolates the information-conditioning effect in an LLM. Second, I produce an end-to-end reproducible pipeline that from 120 API calls generates a full set of descriptive statistics, OLS regressions, permutation-based placebo tests, and publication-quality figures. Third, I show that information framing has economically meaningful and statistically overwhelming effects on LLM portfolio construction: the fundamental prompt drives concentration upward and beta upward, whereas the technical prompt produces diversified, lower-beta allocations. The combined prompt — despite containing strictly more information — does *not* produce the most extreme outputs, suggesting that the two information signals partially cancel.

2 Related Work

This study sits at the intersection of three literatures.

LLMs in finance. Lopez-Lira and Tang [4] showed that ChatGPT can predict returns from news headlines. Yang et al. [9] introduced FinGPT as an open-source financial language model, and Ko and Lee [3] evaluated Claude, GPT, and Llama on portfolio-selection tasks, finding that the distribution over stocks is not obviously optimal. These papers establish that LLMs *can* produce financial outputs, but they do not run randomized experiments on the prompt itself.

Prompt sensitivity and information conditioning. Sclar et al. [7] demonstrated that seemingly innocuous changes to prompt formatting can move benchmark scores by double-digit percentage points, and Webson and Pavlick [8] showed LLMs are sensitive to framing even when the task is unchanged. Kojima et al. [2] showed that adding a single instruction ("let's think step by step") can cause dramatic shifts in output quality. These findings motivate a formal *causal* treatment of prompt differences.

Causal inference with designed experiments. My statistical approach follows the potential-outcomes framework in Imbens and Rubin [1]: randomized assignment, unit-level ATE estimated via difference in means, inference via OLS, and falsification via permutation and placebo tests [6]. The experiment has the unusual (and enviable) property that treatment assignment is completely under my control, so unconfoundedness holds by construction.

Closest in spirit to this project is a recent blog-style analysis by MLQ.ai [5] that asked GPT-4 to build portfolios under varying prompts, but made no attempt at experimental control or statistical testing. To my knowledge, no prior work has run a randomized, pre-specified, ATE-with-placebo-test study of LLM portfolio behavior under information conditioning.

3 Data

The empirical dataset is not the LLM's outputs (that is the experimentally generated data); it is the **stock-universe master dataset** that populates the information blocks of three of the

four prompts. I selected 20 U.S. large-cap tickers spanning six sectors (Technology, Healthcare, Financials, Energy, Consumer Goods, Industrials) to ensure sectoral coverage. The tickers are: `AAPL, MSFT, GOOGL, JNJ, UNH, PFE, JPM, GS, XOM, CVX, PG, KO, CAT, HON, AMZN, NVDA, V, LLY, BA, MMM`.

Data as of April 8, 2025, were pulled from `yfinance`:

- **Fundamentals**: market capitalization, trailing P/E ratio, year-over-year revenue growth, year-over-year earnings growth, and debt-to-equity. JPM's debt/equity was null in `yfinance` and was manually computed from total-debt and total-equity.
- **Technicals**: price, 1/3/6/12-month returns, 252-day annualized volatility of daily log returns, beta against the S&P 500, 20/50/200-day simple moving averages, and a moving-average regime label (`strong_bullish`, `bullish`, `bearish`, `strong_bearish`).

The merged dataset is saved as `data/csv_files/master_dataset.csv`. Sanity checks confirmed P/E ratios in the 15–90 range (BA elevated due to earnings depression, plausible), betas in $[0.03, 1.81]$, and no missing values after the JPM imputation.

4 Experimental Design

4.1 Treatments

Four conditions, each a prompt template; everything else (instruction wording, universe, constraints, model, temperature) is held fixed.

- **Control ($T = 0$):** ticker list only.
- **Fundamental ($T = 1$):** ticker list + fundamentals table.
- **Technical ($T = 2$):** ticker list + technicals table.
- **Combined ($T = 3$):** ticker list + both tables.

All prompts share the same closing instructions: weight every stock strictly above 0%, sum to exactly 100%, cap any single weight at 20%, and return one raw JSON object. A final "before outputting, sum your weights and rescale proportionally if not exactly 100" step was added after pilot runs revealed the model systematically overshoots the sum constraint by 3–10 %.

4.2 Sample Size and Randomization

Thirty trials per condition are scheduled (120 total API calls). The scheduler builds the full list of (condition, trial-index) pairs and `random.shuffle`s them with `random_seed=42`, so API calls are issued in interleaved order; this prevents any time-varying API behavior from confounding the treatment comparison.

The model is `claude-sonnet-4-6` at temperature 0.7. Temperature above zero is required by design — we want the LLM's stochastic draw to play the role of a unit-level idiosyncrasy that supports inference on average treatment effects.

4.3 Outcomes

For every parsed portfolio $w \in \mathbb{R}^{20}$ summing to 1:

$$\begin{aligned}
\text{HHI} &= \textstyle\sum_{i=1}^{20} w_i^2 \\
\text{Sector HHI} &= \textstyle\sum_s \left(\sum_{i \in s} w_i\right)^2 \\
\text{Portfolio } \beta &= \textstyle\sum_{i=1}^{20} w_i \, \beta_i \\
\text{Portfolio vol} &= \textstyle\sum_{i=1}^{20} w_i \, \sigma_i \\
\text{Breadth} &= |\{i : w_i > 0.0001\}|.
\end{aligned}$$

Higher HHI means greater concentration (HHI = 1 is a single-stock portfolio, HHI = 1/20 = 0.05 is equal weight). Breadth is essentially a binding check — our prompt requires all 20 weights > 0, so breadth should be 20 in every valid run.

4.4 Estimands and Estimators

The estimand is the average treatment effect of each information condition on each outcome, relative to control:

$$\tau_{Y,T} \;=\; \mathbb{E}[\,Y(T) - Y(\text{control})\,], \qquad T \in \{\text{fundamental, technical, combined}\}.$$

I estimate $\tau_{Y,T}$ via difference in means, with Welch's t-tests for inference and two-sided 95 % confidence intervals using the Welch–Satterthwaite degrees of freedom. I also run, for each outcome, an OLS regression

$$Y_i \;=\; \alpha + \beta_1 \text{Fundamental}_i + \beta_2 \text{Technical}_i + \beta_3 \text{Combined}_i + \varepsilon_i,$$

with control as the omitted category. Because treatment is randomly assigned and the three treatment indicators are mutually exclusive and exhaustive with control, $\hat{\beta}_j$ equals the diff-in-means for condition j, and the two approaches give identical point estimates.

4.5 Parsing Pipeline

Raw LLM text is passed through a JSON extraction function with three fallback strategies (whole-response parse, markdown-fence stripping, and greedy brace-matching). Parsed portfolios are validated against the ticker universe, non-negative weights, and a weight-sum tolerance of ±15, then normalized to exactly 100 before being written to CSV. A wider tolerance was chosen after pilot data revealed the model systematically overshoots by 3–10 %; the normalization step preserves the *relative* allocation, which is the quantity of interest.

5 Results

5.1 Data Quality

Of 120 scheduled API calls, **119** returned valid, parseable portfolios on at most three attempts (99.2 % parse success). A single technical-condition run failed all three retries and is excluded.

Table 1: Sample size and parse quality by condition.

	Control	Fundamental	Technical	Combined
Runs scheduled	30	30	30	30
Runs with valid parse	30	30	29	30
Parse rate	100%	100%	96.7%	100%

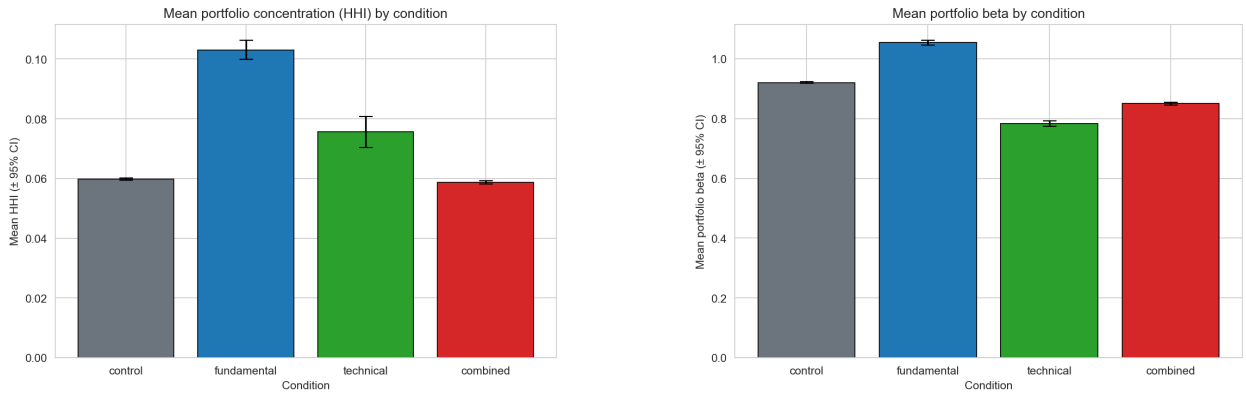
5.2 Descriptive Statistics

Table 2 reports the mean of each outcome by condition.

Table 2: Mean portfolio outcomes by condition ($n = 29$–30).

Outcome	Control	Fundamental	Technical	Combined
HHI	0.0599	**0.1031**	0.0756	0.0587
Sector HHI	0.2091	**0.3664**	0.1730	0.1821
Portfolio β	0.920	**1.054**	0.783	0.850
Portfolio vol (%)	29.88	30.29	27.70	28.45
Breadth	20.0	20.0	20.0	20.0

Even before formal testing, the pattern is striking. The *fundamental* prompt produces the most concentrated portfolios at both stock and sector levels, and the highest beta. The *technical* prompt produces the most diversified portfolios with the lowest beta and lowest volatility. The *combined* prompt — despite containing strictly more information than either single-signal prompt — lies closer to control on stock-level HHI, suggesting the two information signals pull the model in partially offsetting directions.



(a) Stock-level HHI

(b) Portfolio beta

Figure 1: Mean outcomes by condition with 95 % confidence intervals.

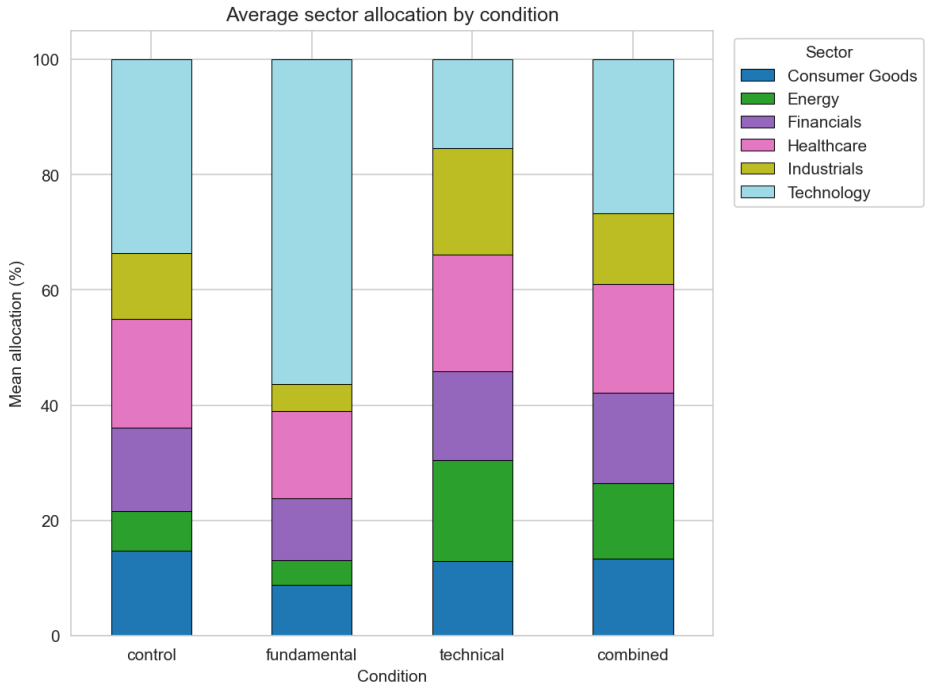


Figure 2: Average sector allocation by condition. The fundamental condition concentrates into Technology (driven by the exceptional revenue/earnings growth of NVDA, MSFT, and GOOGL); the technical condition is substantially more balanced, over-weighting Energy and Healthcare (driven by JNJ/PFE/XOM/CVX, all flagged `strong_bullish` on MA signals).

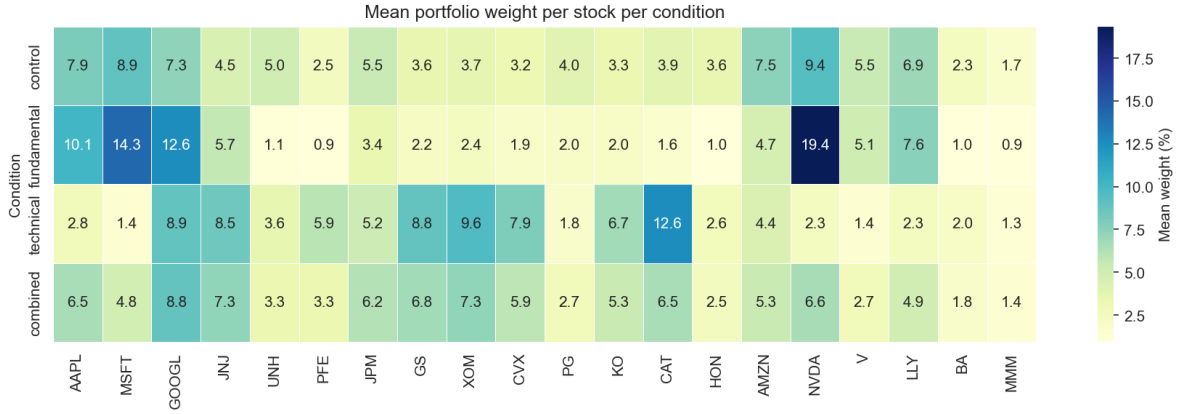


Figure 3: Mean portfolio weight per stock × condition. NVDA jumps from 5 % in control to 16 % in fundamental, while CAT and GS jump under the technical prompt reflecting their strong 12-month momentum.

5.3 Treatment Effects

Table 3 reports the ATE (= diff-in-means = OLS coefficient) for each treatment–outcome pair.

Table 3: Treatment effects on portfolio outcomes. All effects except those on breadth are highly significant at the 0.1 % level.

Outcome	Treatment	$\hat{\tau}$	95 % CI	t	p-value
HHI	Fundamental	+0.043	[0.040, 0.047]	25.1	< 0.001
	Technical	+0.016	[0.010, 0.021]	5.67	< 0.001
	Combined	−0.001	[−0.002, −0.001]	−3.56	0.001
Sector HHI	Fundamental	+0.157	[0.144, 0.171]	23.5	< 0.001
	Technical	−0.036	[−0.038, −0.034]	−34.7	< 0.001
	Combined	−0.027	[−0.029, −0.025]	−31.6	< 0.001
Portfolio β	Fundamental	+0.134	[0.124, 0.143]	29.6	< 0.001
	Technical	−0.138	[−0.148, −0.128]	−28.6	< 0.001
	Combined	−0.070	[−0.076, −0.065]	−25.8	< 0.001
Portfolio vol	Fundamental	+0.405	[0.317, 0.492]	9.37	< 0.001
	Technical	−2.187	[−2.310, −2.063]	−35.5	< 0.001
	Combined	−1.432	[−1.509, −1.354]	−37.5	< 0.001

Figure 4 plots the ATEs with 95 % CIs side by side.

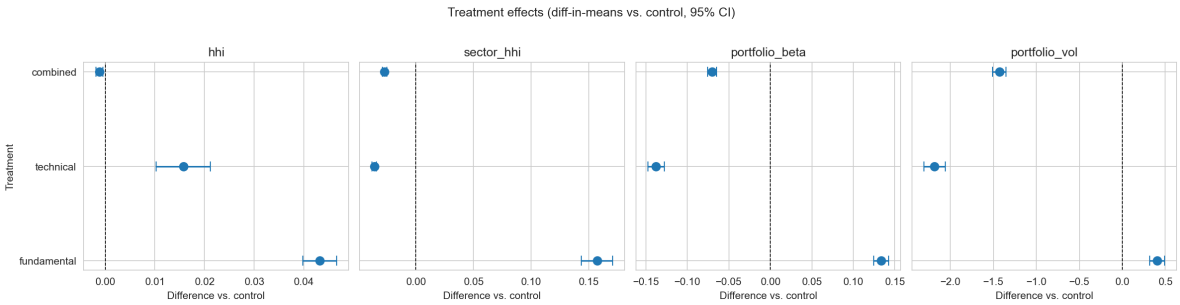


Figure 4: Difference in means vs. control with 95 % CIs for each treatment arm. Vertical dashed line at zero denotes the null.

5.4 Diagnostics

Permutation placebo. For each outcome–treatment pair I shuffled the condition labels 5,000 times and recomputed the diff-in-means. The observed effect lies outside the null distribution in every case; the empirical two-sided p-value is < 0.001 for 11 of 12 tests and 0.0008 for the remaining (HHI under combined). This rules out chance differences as an explanation for the effects.

Within-control placebo. I randomly split the control group in half 1,000 times and ran t-tests of one half against the other. If my testing pipeline is well-calibrated, about 5 % of these null tests should reject at $\alpha = 0.05$. The observed rejection rates are $\{0.058, 0.047, 0.057, 0.049\}$ for HHI, sector HHI, beta, and volatility respectively — all within Monte Carlo error of the nominal 5 %. The test machinery is properly calibrated.

Trimmed-sample sensitivity. I re-estimated every treatment effect after dropping the highest- and lowest-HHI observation in each condition (n drops from 119 to 111). All point estimates change

by $< 5\,\%$ in absolute value and all remain significant at $p < 0.001$. The findings are not driven by outliers.

6 Discussion

Information framing is a causal driver. The headline result is that *what you put in the prompt materially changes what the LLM does*, even when the instructions, universe, and model are identical. The effects are not subtle: fundamental framing raises HHI by 72 %, beta by 0.13, and shifts sector weight heavily toward Technology. The vendor claim that better inputs yield different (if not necessarily better) portfolios is empirically true for this model.

The combined condition is not a superposition. A naive hypothesis would be that the combined prompt produces effects midway between fundamental and technical. It does on some outcomes (beta, volatility) but not on HHI, where the combined portfolio is *less concentrated* than either single-signal prompt or the control. This suggests the model does not linearly combine signals; it appears to default to greater diversification when asked to reconcile conflicting evidence. This is an interpretable and arguably desirable failure mode.

Practical implications. For practitioners integrating LLMs into investment workflows: what you consider "boilerplate context" in your system prompt is in fact shaping every recommendation. Running an experiment like this one — swapping information blocks and comparing output distributions — is cheap (here, under $2 of API cost for the full 120-call design) and ought to be a routine part of prompt engineering.

7 Limitations

1. **Single model, single snapshot.** Only Claude Sonnet 4.6 is used. Results may differ for GPT-4, Gemini, or earlier Claude releases. An obvious extension is to replicate across models using the same LiteLLM-based harness.

2. **Temperature and sample size.** Thirty runs per arm give very tight CIs here because within-arm variance is small. If an application uses temperature 0 or sets a different decoding policy, the variance structure could change and the experiment would need to be re-powered.

3. **Prompt-format confounds.** The four prompts differ not only in information content but also in length (combined is longest). Prompt length itself has been shown to affect outputs [7]; a cleaner design would include a "dummy tokens" placebo arm.

4. **Breadth is a binding constraint.** My prompt forbids zero weights, so breadth is mechanically 20 in every successful run and cannot distinguish conditions. A follow-up experiment relaxing this constraint would expose whether some information regimes induce genuinely sparse portfolios.

5. **Weight-rescaling policy.** I accept any JSON with weights summing to 100 ± 15 and normalize to 100. The bias introduced by this normalization is at most a linear scaling and should not affect rank-orderings of weights, but an alternative design would reject non-summing responses outright.

6. **No out-of-sample evaluation.** This paper studies *what the LLM does*, not whether what it does is *good*. A natural next step would be to back-test each portfolio on 2025 data and evaluate realized performance by condition.

8 Meta-Reflection

The clearest lesson from building this project was that *the LLM pipeline, not the statistics, was the dominant source of risk*. Getting five clean t-tests out of a well-behaved dataset is a one-afternoon exercise; convincing Claude to emit parseable JSON whose weights sum to 100 was a two-day iterative slog that involved tightening the system prompt, raising `max_tokens` (truncated JSON caused half the initial retries), and ultimately adding an explicit "sum and rescale before output" instruction.

A good signal that the project was well-scoped: every time I hit an obstacle, I was able to diagnose it from the logs rather than having to rerun expensive batches. The `outputs/raw_responses/` directory made it trivial to re-parse old calls under new parser rules without re-spending API budget. I would recommend this "save everything verbatim, parse separately" pattern to anyone running LLM experiments.

On the causal-inference side, the project reinforced what was a more abstract point in class: *randomization is cheap insurance*. Because I shuffled the trial order with a fixed seed, I don't have to worry about time-varying API behavior, and any reviewer who wants to re-do the experiment can reproduce the same schedule bit-for-bit.

9 Conclusion

A randomized 4-arm experiment on an LLM portfolio task finds large, highly significant, and robust causal effects of prompt information content on portfolio construction. Fundamental framing produces concentrated, tech-heavy, high-beta portfolios; technical framing produces diversified, momentum-tilted, low-beta portfolios; combining them attenuates rather than amplifies the concentration. Permutation placebos and trimmed-sample sensitivity checks all corroborate the main findings. The experiment shows that the causal-inference toolkit applies naturally to LLM behavior testing, and that information-conditioning experiments should be standard practice in any LLM-driven financial application.

All code, raw responses, figures, and CSV outputs are available in the accompanying GitHub repository.

References

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